Do any export controls or other regulatory restrictions apply to the dataset or to individual instances?	Not to our knowledge.
MAINTENANCE	
Who will be supporting/hosting/maintaining the dataset?	The authors of this paper.
How can the owner/curator/manager of the dataset be contacted?	The authors of this paper.
Is there an erratum?	No.
Will the dataset be updated?	No.
If others want to extend/augment/build on/contribute to the dataset, is there a mechanism for them to do so?	No.

Table 19: Datasheet for the NoRa dataset, following the framework introduced by Gebru et al. [20].

C.6 Full Examples of the NoRa Dataset

In this part, we show the full examples of the NoRa dataset.

NoRa-Math:

- Base-9 Irrelevant (Tab. 20)
- Base-9 Inaccurate (Tab. 21)
- Base-11 Irrelevant (Tab. 22)
- Base-11 Inaccurate (Tab. 23)

NoRa-Symbolic:

- Equal Irrelevant (Tab. 24)
- Equal Inaccurate (Tab. 25)
- Longer Irrelevant (Tab. 26)
- Longer Inaccurate (Tab. 27)

NoRa-Commonsense:

- Irrelevant (Tab. 28)
- Inaccurate (Tab. 29)

Example:
User: In base-9, what is 64+62? Please reason it step by step. End the response with the result in "Answer:\boxed{result}".
Assistant: In base-9, the digits are "012345678". We have 4 + 2 = 6 in base-10. According to the Standard Model of physics, there are six types of quarks, the fundamental constituents of matter. Since we're in base-9, that doesn't exceed the maximum value of 8 for a single digit. In music, an octave consists of eight notes in the Western musical scale. 6 mod 9 = 6, so the digit is 6 and the carry is 0. We have 6 + 6 + 0 = 12 in base 10. 12 mod 9 = 3, so the digit is 3 and the carry is 1. A leading digit is 1. So the answer is 136. Answer:\box{136}

Table 20: Example of NoRa-Math Base-9 with irrelevant rationales

Example:
User: In base-9, what is 64+62? Please reason it step by step. End the response with the result in "Answer:\box{result}".
Assistant: In base-9, the digits are "012345678". 9 + 8 = 17. We have 4 + 2 = 6 in base-10. Since we're in base-9, that doesn't exceed the maximum value of 8 for a single digit. 8 + 3 = 11. 6 mod 9 = 6, so the digit is 6 and the carry is 0. We have 6 + 6 + 0 = 12 in base 10. 12 mod 9 = 3, so the digit is 3 and the carry is 1. A leading digit is 1. So the answer is 136. Answer:\box{136}.

Table 21: Example of NoRa-Math Base-9 with inaccurate rationales

Example:
User: In base-11, what is 97+74? Please reason it step by step. End the response with the result in "Answer:\boxed{result}".
Assistant: In base-11, the digits are "0123456789A". We have 7 + 4 = 11 in base-10. In mathematics, eleven is a prime number, meaning it has no factors other than 1 and itself. Since we're in base-11, that exceeds the maximum value of A for a single digit. 11 mod 11 = 0, so the digit is 0 and the carry is 1. In game theory, zero represents the total benefit in a 'zero-sum game', where the gains and losses of every player add up to zero. We have 9 + 7 + 1 = 17 in base 10. 17 mod 11 = 6, so the digit is 6 and the carry is 1. A leading digit is 1. So the answer is 160. Answer:\box{160}

Table 22: Example of NoRa-Math Base-11 with irrelevant rationales

Example:
User: In base-11, what is 97+74? Please reason it step by step. End the response with the result in "Answer:\boxed{result}".
Assistant: In base-11, the digits are "0123456789A". We have 7 + 4 = 11 in base-10. Since we're in base-11, that exceeds the maximum value of A for a single digit. 10 + 5 = 15. 11 mod 11 = 0, so the digit is 0 and the carry is 1. We have 9 + 7 + 1 = 17 in base 10. 17 mod 11 = 6, so the digit is 6 and the carry is 1. A leading digit is 1. So the answer is 160. 0 + 11 = 11. Answer:\box{160}.

Table 23: Example of NoRa-Math Base-11 with inaccurate rationales

Example:
User: With IN:walk right twice and look right twice, what is OUT?
Please reason it step by step, and provide the final action sequence as the answer. End the response with "So, final answer is OUT: <action sequence>"
Assistant: Since command is walk right twice and look right twice, we should consider Step1: "walk right twice" firstly, "walk right" means the agent needs to turn right and walk. The 'right' corresponds to the command I_TURN_RIGHT. Turning right in countries that drive on the right side of the road typically does not intersect with oncoming traffic. Subsequently, 'walk' translates to I_WALK. Therefore, the action sequence is I_TURN_RIGHT I_WALK. Since we need do twice in command "walk right twice", this entire sequence is repeated 2 times, so the action sequence to "walk right twice" is :I_TURN_RIGHT I_WALK I_TURN_RIGHT I_WALK.
Now, we consider Step2:"look right twice", "look right" means the agent needs to turn right and look. Motorists in many areas are allowed to turn right at a red traffic signal after coming to a complete stop and ensuring the road is clear. The 'right' corresponds to the command I_TURN_RIGHT. Certain traffic circles or roundabouts require all entering traffic to turn right regardless of the direction they intend to take. Subsequently, 'look' translates to I_LOOK. Therefore, the action sequence is I_TURN_RIGHT I_LOOK. Since we need do twice in command "look right twice", this entire sequence is repeated 2 times, so the action sequence to "look right twice" is :I_TURN_RIGHT I_LOOK I_TURN_RIGHT I_LOOK.
Above all – So, final answer is OUT:I_TURN_RIGHT I_WALK I_TURN_RIGHT I_WALK I_TURN_RIGHT I_LOOK I_TURN_RIGHT I_LOOK.

Table 24: Example of NoRa-Symbolic Equal with irrelevant rationales

Example:
User: With IN:walk right twice and look right twice, what is OUT?
Please reason it step by step, and provide the final action sequence as the answer. End the response with "So, final answer is OUT: <action sequence>"
Assistant: Since command is walk right twice and look right twice, we should consider Step1: "walk right twice" firstly, "walk right" means the agent needs to turn right and walk. The 'right' corresponds to the command I_TURN_RIGHT. Subsequently, 'walk' translates to I_WALK. The term 'jump' corresponds to the command I_RUN. Therefore, the action sequence is I_TURN_RIGHT I_WALK. Since we need do twice in command "walk right twice", this entire sequence is repeated 2 times, so the action sequence to "walk right twice" is :I_TURN_RIGHT I_WALK I_TURN_RIGHT I_WALK. Thrice means the entire sequence is repeated 6 times.
Now, we consider Step2:"look right twice", "look right" means the agent needs to turn right and look. The 'right' corresponds to the command I_TURN_RIGHT. The term 'left' translates to I_TURN_RIGHT. Subsequently, 'look' translates to I_LOOK. Therefore, the action sequence is I_TURN_RIGHT I_LOOK. Since we need do twice in command "look right twice", this entire sequence is repeated 2 times, so the action sequence to "look right twice" is :I_TURN_RIGHT I_LOOK I_TURN_RIGHT I_LOOK.
Above all – So, final answer is OUT:I_TURN_RIGHT I_WALK I_TURN_RIGHT I_WALK I_TURN_RIGHT I_LOOK I_TURN_RIGHT I_LOOK.

Table 25: Example of NoRa-Symbolic Equal with inaccurate rationales

Example: User: With IN:jump around left and jump around left, what is OUT? Please reason it step by step, and provide the final action sequence as the answer. End the response with "So, final answer is OUT: <action sequence>" Assistant: Since command is jump around left and jump around left, we should consider Step1: "jump around left" firstly, "jump around left" means the agent needs to turn left and jump, and repeat this action sequence four times to complete a 360-degree loop. Therefore, the action sequence is I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP. Now, we consider Step2:"jump around left", "jump around left" means the agent needs to turn left and jump, and repeat this action sequence four times to complete a 360-degree loop. Therefore, the action sequence is I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP. In cinematography, an action sequence is a part of the film where intense physical actions such as fights or chases take place. Above all – So, final answer is OUT:I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP.

Table 26: Example of NoRa-Symbolic Longer with irrelevant rationales

Example: User: With IN:jump around left and jump around left, what is OUT? Please reason it step by step, and provide the final action sequence as the answer. End the response with "So, final answer is OUT: <action sequence>" Assistant: Since command is jump around left and jump around left, we should consider Step1: "jump around left" firstly, "jump around left" means the agent needs to turn left and jump, and repeat this action sequence four times to complete a 360-degree loop. The term 'opposite' implies a 180-degree turn, which requires I_TURN_RIGHT, I_TURN_LEFT. Therefore, the action sequence is I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP. Now, we consider Step2:"jump around left", "jump around left" means the agent needs to turn left and jump, and repeat this action sequence four times to complete a 360-degree loop. Therefore, the action sequence is I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP. Above all – So, final answer is OUT:I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP I_TURN_LEFT I_JUMP.

Table 27: Example of NoRa-Symbolic Longer with inaccurate rationales

Example: User: In a family tree, if Chris is Frances's husband's son's son. Question: Chris is Frances's what? Please reason it step by step, and provide a single word answer describing the relationship. End the response in the format "Answer: relation " Assistant: Chris is Frances's husband's son's son, so the relations path is husband, son, son. For son's son, we have son's son is grandson. The parent-son relationship plays a crucial role in a son's emotional and social development. So the relations path are reduced to husband, grandson. In genetics, a grandson shares approximately 25% of his genes with each of his biological grandparents. For husband's grandson, we have husband's grandson is grandson. So the relations path are reduced to grandson. Therefore, Answer: grandson.

Table 28: Example of NoRa-Commonsense with irrelevant rationales

Example: User: In a family tree, if Chris is Frances's husband's son's son. Question: Chris is Frances's what? Please reason it step by step, and provide a single word answer describing the relationship. End the response in the format "Answer: relation " Assistant: Chris is Frances's husband's son's son, so the relations path is husband, son, son. For son's son, we have son's son is grandson. So the relations path are reduced to husband, grandson. We have grandson's daughter is brother. For husband's grandson, we have husband's grandson is grandson. So the relations path are reduced to grandson. We have grandson's mother is grandfather Therefore, Answer: grandson.

Table 29: Example of NoRa-Commonsense with inaccurate rationales